PHYSICS

SECTION – I
Single Correct Choice Type

This section contains 4 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

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1. A small, hollow metal cylinder is closed at one end. Its mass is known which of the following instruments are required to find the density of the metal?

(a) slide callipers only (b) slide callipers and screw gauge

(c) screw gauge and spherometer (d) slide callipers and spherometer

2. Protons accelerated through a potential difference V travels in the form of beams of radius r and constitutes a current *I*. The electric field at the surface of the beams is (take e as charge of proton and mass of the proton is m)

(a) $\frac{I}{2\pi r\varepsilon_0}\sqrt{\frac{m}{2\,\text{eV}}}$ (b) $\frac{I}{4\pi r\varepsilon_0}\sqrt{\frac{m}{2\,\text{eV}}}$ (c) $\frac{I}{2\pi r\varepsilon_0}\sqrt{\frac{m}{\text{eV}}}$ (d) $\frac{I}{2\pi r\varepsilon_0}\sqrt{\frac{m}{4\,\text{eV}}}$

3. Let us assume that the temperature of the air varies with height linearly from the value T_1 = 300 K at y_1 = 0 to T_2 = 250 K at y_2 = 10 km. The time *t* needed for a sound wave produced at the earth's surface to reach the height y_2 is (velocity of sound at 0^0C = 330 m/s)

(a) 30 s (b) 60 s (c) 15 s (d) 90 s

4. The internal energy of an imaginary gas is determined by the equation

$$U = a\ln\left(\frac{T}{T_0}\right) + b\ln\left(\frac{P}{P_0}\right)$$

Where a = 3 kJ, b = 7 kJ, T_0 = 200 K and P_0 = 10 KPa

On importing 500 J of heat of constant pressure of P = 10^5 Pa the volume changes (increase) by 0.5×10^{-3} m^3. If final temperature is *n* times the initial temperature then *n* is

(a) 1.00 (b) 1.17 (c) 2.00 (d) 2.17

SECTION – II
Multiple Correct Choice Type

This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

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5. A capacitor of capacity C has charge q_0 in fully charged condition. The capacitor is now connected to a coil of inductance L. Which of following are correct?

(a) at an instant $t = T/8$, charge $= \frac{q_0}{\sqrt{2}}$

(b) magnetic energy at $t = \frac{T}{8}$ is 50% of the total energy

(c) ratio of magnetic energy to that of electorate energy is $\frac{1}{2}$ at any instant

(d) ratio of electrostatic energy to magnetic energy is 1 only at q is 70.7 % of maximum charge.

6. A sphere of radius R uniformly charged with surface charge density σ, rotates around the axis passing through its centre at an angular velocity ω. Now

(a) magnetic field at centre $= \frac{2}{3}\mu_0\omega\sigma R$

(b) magnetic moment $= \frac{4}{3}\pi\sigma R^4\omega$

(c) magnetic field at centre $= \frac{1}{3}\mu_0\omega\sigma R$

(d) magnetic moment $= \frac{2}{3}\pi\sigma R^4\omega$

7. A sphere of mass m = 200 gm fixed between 2 springs of total force constant k = 28.8 N/m can slide over a frictionless platform PQ placed horizontally. The system rotates with an angular velocity ω = 8 rad/sec about a vertical axis OO'. Now,

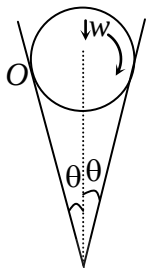
(a) time period of system is 0.7 sec and

(b) time period of system is 0.4 sec and for ω = 12 rad/sec, it stops

(c) time period of system is 0.7 sec and for ω = 12 rad/sec, the oscillator will stop

(d) time period of system is 0.4 sec and for all value of ω, the oscillation continues

8. A rotor of weight w and radius r rests in a v-shaped support as shown. The coefficient of friction at the contact is μ. The couple required to produce rotation is



(a) $\frac{\mu wr}{(1+\mu^2)\sin\theta}$ maintaining the contact O and wr cos θ in the absence of contact O

(b) $\frac{\mu wr}{(1+\mu^2)\sin\theta}$ when in contact at O is maintained

(c) wr cos θ in the absence of contact O

(d) $\frac{2\mu wr}{(1+\mu^2)\cos\theta}$ maintaining the contact O and wr cos θ in the absence of contact O.

SECTION – III
Reasoning Type

This section contains 4 reasoning type questions. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True; **Statement-2** is a correct explanation for **Statement-1**.

(B) **Statement-1** is True, **Statement-2** is True; **Statement-2** is not a correct explanation for **Statement-1**.

(C) **Statement-1** is True, **Statement-2** is False.

(D) **Statement-1** is False, **Statement-2** is True.

9. **Statement-1:** The rate of change of momentum of a many particle system is proportional to the vector sum of the internal forces.

Statement-2: Internal forces can change the kinetic energy as well as momentum of the body.

(a) A (b) B (c) C (d) D

10. **Statement-1:** If the sum of the 2 unit vectors is also a unit vector, then magnitude of their difference is square root of three.

Statement-2: To find resultant of 2 vectors, we use triangle law.

(a) A (b) B (c) C (d) D

11. **Statement-1:** The images formed by total internal reflections are much brighter than those formed by mirrors or lens.

Statement-2: There is no lose of intensity in total internal reflections.

(a) A (b) B (c) C (d) D

12. **Statement-1:** If the frequency of the light incident on a metal surface is halved, the maximum K.E. of emitted photoelectrons also gets halved.

Statement-2: K.E. of a particle is directly proportional to frequency.

(a) A (b) B (c) C (d) D

SECTION–IV
Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Passage-I

A resistor R and an inductor L are connected in series to a battery of volt (V) with zero internal resistance. Due to self inductor an induced emf is setup across L. Magnetic energy is stored in an inductor. Use R = 5Ω L = 500 H V = 10 V

13. What is the current in the circuit when the rate of increase of energy in inductor is maximum

(a) 1 amp (b) 2 amp (c) 3 amp (d) 4 amp

14. What is the maximum current in the circuit?

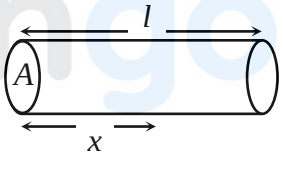
(a) 1 amp (b) 2 amp (c) 3 amp (d) 4 amp

15. When will the joule heat dissipated in the resistor change at the fastest rate?

(a) 69 sec (b) 34 sec (c) 138 sec (d) 40 sec

Passage-II

The cross sectional area and length of a cylindrical conductor are A and l respectively. The specified conductivity varies as $\sigma = \sigma_0 \frac{l}{\sqrt{x}}$, when x is the distance along the axis of the cylinder from one of its ends.



16. What is the resistance of cylinder along its cylindrical axis?

(a) $\frac{3\sqrt{l}}{2A\sigma_0}$ (b) $\frac{\sqrt{l}}{3A\sigma_0}$ (c) $\frac{2\sqrt{l}}{A\sigma_0}$ (d) $\frac{2\sqrt{l}}{3A\sigma_0}$

17. What is the current density it potential does along the cylinder v_0?

(a) $\frac{3\sigma_0 v_0}{2\sqrt{l}}$ (b) $\frac{2\sigma_0 v_0}{3\sqrt{l}}$ (c) $\frac{\sigma_0 v_0}{2\sqrt{l}}$ (d) $\frac{3\sigma_0 v_0}{\sqrt{l}}$

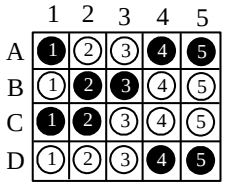
18. The electric field at each point in the cylinder in this case is

(a) $\frac{3v_0\sqrt{x}}{2(l)^{3/2}}$ (b) $\frac{2v_0\sqrt{x}}{3(l)^{3/2}}$ (c) $\frac{v_0\sqrt{x}}{(l)^{3/2}}$ (d) $\frac{2v_0\sqrt{x}}{(l)^{3/2}}$

SECTION–V
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.

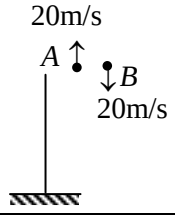
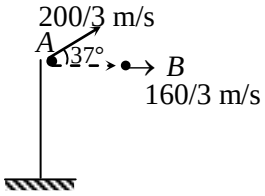
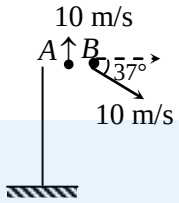
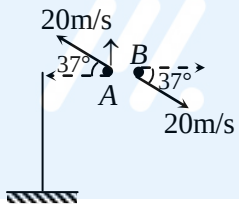


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1 Some situations are given with reference to Young's double slit experiment in column I and their effects are mentioned in column II. Match these situations with their corresponding effects.

Column I		Column II	
(A)	Separation between screen and slits is increased	**1.**	fringe width increases
(B)	Slit separation is increases symmetrically	**2.**	fringe width decreases
(C)	The whole arrangement is immersed in water	**3.**	central maxima shifts either up or down
(D)	A thin slab is introduced in front of any one of the source	**4.**	all the fringes shifts by the same distance
		5.	all the fringes shifts by different distances

2. Both *A* and *B* are thrown simultaneously as shown from a high tower. Match column - I with Column - II

Column - I	**Column -II**
(A) 	**1.** Distance between the two balls after 2 seconds is $16\sqrt{5}$ m.
(B) 	**2.** Distance between the two balls after 2 seconds is 80 m.
(C) 	**3.** Magnitude of relative velocity of *B* with respect to *A* is $8\sqrt{5}$ m/s.
(D) 	**4.** Magnitude of relative velocity of *B* with respect to *A* is 40 m/s. **5.** Magnitude of relative horizontal separation of A with respect to B is zero.

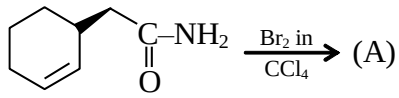
CHEMISTRY

SECTION – I

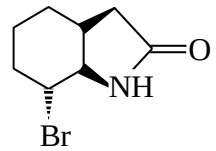
Single Correct Choice Type

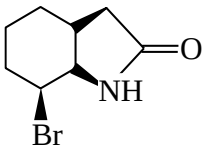
This section contains 8 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

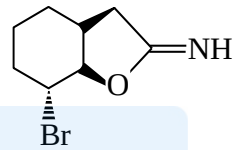
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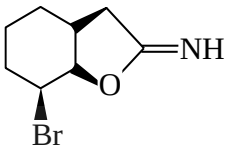
1. 

The product (A) in the given reaction is

(a) 

(b) 

(c) 

(d) 

2. A solution of a monoprotic weak acid has acidity constant K_a. The minimum initial concentration 'c' in terms of K_a, such that the concentration of the undissociated acid can be equated to 'c' within a 10% limit of error, would be

(a) 30 K_a (b) 50 K_a

(c) 70 K_a (d) 90 K_a

3. An optically active substance undergoes first order mutarotation in an aqueous solution. The observed rotation at 0 min, 180 min and ∞ time are 190°, 150° and 30° respectively. The sum of forward and backward rate constants ($k_f + k_r$) would be (log 3 = 0.477, log 2 = 0.3)

(a) 1.57×10^{3} min^{-1} (b) 1.57×10^{-2} min^{-1}

(c) 1.57×10^{-3} min^{-1} (d) 15.7 min^{-1}

4. Which of the following is an incorrect statement? (Atomic numbers: P = 15, S = 16, I = 53)

(a) Axial P–Cl bonds in PCl_5 are longer than equatorial P–Cl bonds.
(b) SF_4 has distorted tetrahedral arrangement.
(c) Bond order of I–I bond in I_3^- is one.
(d) Naphthalene has two types of C–C bond lengths.

5. Which of the following complex represents correct property indicated with it? (Given: atomic number of Fe = 26, Ni = 28)

(a) $[Fe(CO)_5]$: square pyramidal
(b) $[Ni(DMG)_2]$: Colourless
(c) $[Ni(CN)_4]^{2-}$: $\mu = \sqrt{8}$ BM
(d) $[Fe(CN)_5NO]^{2-}$: d^2sp^3 hybridization

6. 0.2063 g of an organic compound (molar mass 168) was heated with sufficient amount of HI and the resulting solution was treated with alcoholic $AgNO_3$ solution. This led to precipitation of 0.8658 g of AgI. The number of methoxy groups in one molecule of the organic compound is/are (Given: atomic mass of Ag = 108, I = 127)

(a) 1
(b) 2
(c) 3
(d) 4

7. Which of the following metal carbide on hydrolysis gives methane?

(a) Mg_2C_3
(b) SiC
(c) Mn_3C
(d) Na_2C_2

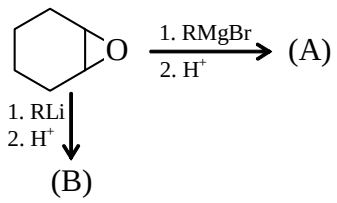
8. Analysis show that nickel oxide consist of nickel ion with 96% ions having d^8 configuration and 4% ions having d^7 configuration. Which among the following best represents the formula of the nickel oxide? (Atomic number of Ni = 28)

(a) $Ni_{1.02}O$
(b) $Ni_{0.96}O$
(c) $Ni_{0.97}O$
(d) $Ni_{0.98}O$

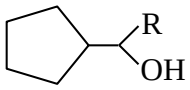
SECTION – II
Multiple Correct Choice Type

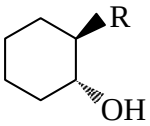
This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

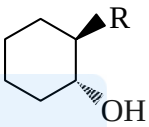
9.

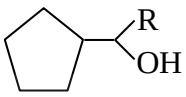


Which of the following is/are correct about structures of compound (A) and (B)?

(a) (A): 

(b) (B): 

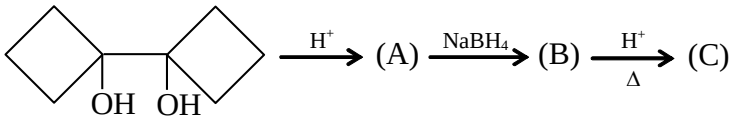
(c) (A): 

(d) (B): 

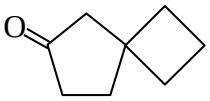
10. Which of the following statement(s) is/are correct for the given reversible process in a state of equilibrium?

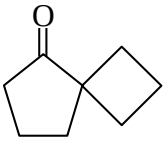
$$2Ag^{+} + 2Cl^{-} \rightleftharpoons Cl_2 + 2Ag$$

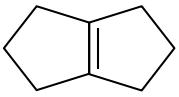
(a) $\Delta G = -2.303\ RT \log K_p$
(b) $\Delta G^\circ = -2.303\ RT \log K_p$
(c) $\Delta G = -2.303\ RT \log K_c$
(d) $\Delta G^\circ = -2.303\ RT \log K_c$

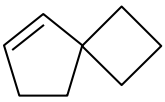
11. 

Which of the following is correct about the products (A), (B) and (C)?

(a) (A): 

(b) (A): 

(c) (C): 

(d) (C): 

12. Two liquids A and B have same molar mass, form a liquid mixture with $X_A = \frac{1}{3}$ and total pressure of 600 mm Hg. The liquid mixture is distilled at 120°C, when $\frac{1}{4}$ of it is distilled. The mole fraction of A in the distillate is 0.75. Which of the following is/are true regarding this mixture? (Given: P_A^o = 300 mm Hg)

(a) Vapour pressure of pure B is 750 mm Hg.

(b) Number of moles of A in the distillate is $\frac{3}{16}$ of the total moles in original solution.

(c) Moles of B in the residue is $\frac{7}{36}$ of the total moles in original solution.

(d) Total vapour pressure of the residue is 662.5 mm Hg.

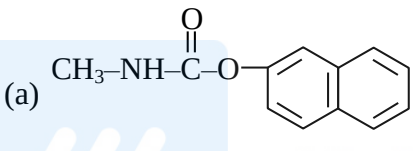
SECTION–III
Comprehension Type

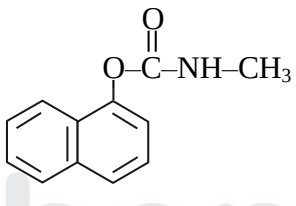
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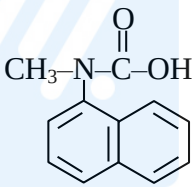
Passage–I

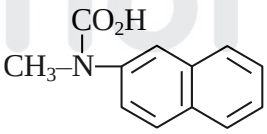
An insecticide SEVIN is made by the reaction of 1-naphthol with methyl isocyanate. A union carbide plant in Bhopal, India, were using this process to make SEVIN for use as an agricultural insecticide. On December 3, 1984, either by accident or by sabotage, a valve was opened that admitted water to a large tank of methyl isocyanate. The pressure and temperatures within the tank rose dramaticity and the pressure relief valves opened to prevent the tank from bursting. A large quantity of methyl isocyanate rushed through the pressure relief valves and the vapours flowed with the breeze into populated areas, killing about 2500 people and injuring many more.

13. The structure of SEVIN is

(a) 

(b) 

(c) 

(d) 

14. The pressure rise in the container (tank) was due to liberation of

(a) NH_3 (b) CH_4

(c) CO_2 (d) CO

15. The temperature rise in the tank was attributed to

(a) Dissolution of NH_3 in water. (b) Dissolution of CO_2 in water.

(c) Dissolution of CH_3–N=C=O in water. (d) Dissolution of CH_3NH_2 in water.

Passage–II

Successive effusion is used for isotopic separation from a gaseous mixture. Let there are two (isotopic) gases A and B present in a container having n_{A_o} and n_{B_o} moles respectively. Effusing this mixture into another container enriches the mixture in that gas which effuses faster. Let, after the first effusion stage the number of moles of A and B are n_{A_1} and n_{B_1} respectively. If there are 'm' number of effusion stages used and finally the number of moles of A and B in the last container is n_{A_m} and n_{B_m} respectively.

The ratio of relative concentration of given (isotopic) species after processing to its relative concentration before processing is referred as separation factor (f).

$$f = \frac{n_{A_m} / n_{B_m}}{n_{A_o} / n_{B_o}}$$

For a single stage effusion process, the maximum (ideal) separation factor (f') is the ratio of rate of effusion of isotopic molecules to the ratio of its relative concentration before processing. This can be determined using Graham's law.

$$f' = \frac{n_{A_1} / n_{B_1}}{n_{A_o} / n_{B_o}} = \frac{r_A / r_B}{n_{A_o} / n_{B_o}}$$

16. Natural uranium consists of 0.1% of U^{235} by mole. The uranium ore is reacted with fluorine to form volatile uranium hexafluoride. The number of effusion steps required to achieve a concentration of 99.9% of $^{235}UF_6$, would be (atomic mass of F = 19)

[Given: log 352 = 2.546, log 349 = 2.542]

(a) 300 (b) 3,000

(c) 30,000 (d) 3,00,000

17. The number of effusion stages needed to enrich initial molar ratio of H_2 and He from 2 : 1 to 64 : 1 by means of successive effusion, would be

(a) 4 (b) 8

(c) 10 (d) 14

18. A sample of hydrogen contains 90% of D_2 and rest is H_2. This mixture is subjected to isotopic enrichment by successive effusion technique. The percentage of H_2 in the mixture after 18 effusion steps is approximately

(a) 100 % (b) 98 %

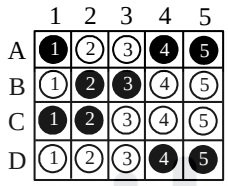
(c) 96 % (d) 93 %

SECTION–IV

Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

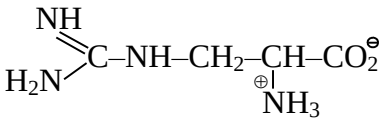
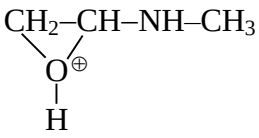
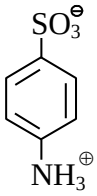
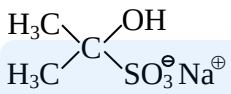
If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



1.

Column I	Column II
(A) $[Co(NO_2)_2(H_2O)_2(NH_3)_2]NO_3$	**1.** Number of stereoisomers = 6
(B) $[Ni(en)_3]Br_3$	**2.** Linkage isomerism
(C) $[Co(NH_3)_3(py)_3]Br_2$	**3.** Ionization isomerism
(D) $[Pt(en)(SCN)_2]\ (NO_3)_2$	**4.** Optical isomerism
	5. Geometrical isomerism

2.

Column I	Column II
(A) 	**1.** Show acidic behaviour towards water.
(B) 	**2.** Show basic behaviour towards water.
(C) 	**3.** Show resonance
(D) 	**4.** Zwitter ion **5.** Crystalline salt

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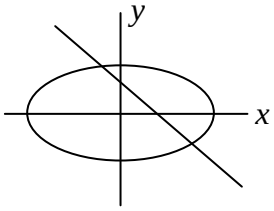
MATHEMATICS

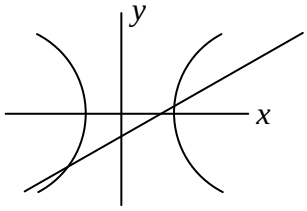
SECTION – I
Single Correct Choice Type

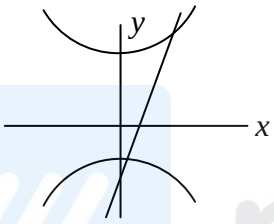
This section contains 4 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

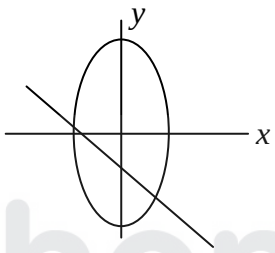
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1. Let $a, b \in R$ where $ab \neq 0$, then the graph of the straight line $ax - y + b = 0$ and the conic section $bx^2 + ay^2 = ab$ can be

(a) 

(b) 

(c) 

(d) 

2. $f : R \to R,\ f(x) = \dfrac{x^2 + bx + 1}{x^2 + 2x + b}$, if function $f(x)$ and $\dfrac{1}{f(x)}$ have same bounded set as its range, then value of b is

(a) $2\sqrt{3} - 2$ (b) $2\sqrt{3} + 2$ (c) $2\sqrt{2} - 2$ (d) $2\sqrt{2} + 2$

3. If the differential equation of a curve, passing through $\left(0, -\dfrac{\pi}{4}\right)$ and $(t, 0)$ is $\cos y\left(\dfrac{dy}{dx} + e^{-x}\right) + \sin y\left(e^{-x} - \dfrac{dy}{dx}\right) = e^{e^{-x}}$, then the value of $t.e^{e^{-t}}$ is

(a) -1 (b) 1 (c) 2 (d) -2

4. The number of positive integral solution of the equation $\begin{vmatrix} y^3+1 & y^2z & y^2x \\ yz^2 & z^3+1 & z^2x \\ yx^2 & x^2z & x^3+1 \end{vmatrix} = 11$ is

(a) 1 (b) 2 (c) 3 (d) 4

SECTION – II
Multiple Correct Choice Type

This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

---

5. Let z_1, z_2, z_3 be the coordinate of vertices A, B, C of the triangle if $W_1 = z_1 - z_2$, $W_2 = z_3 - z_2$ and $\angle A = 90°$, then

(a) $\text{Re}\left(\frac{W_2}{W_1}\right) = 1$ (b) $\text{Re}(W_1\overline{W_2}) = |W_1|^2$

(c) $\text{Re}(W_1\overline{W_2}) = |W_2|^2$ (d) $\text{Re}\left(\frac{W_1}{W_2}\right) = 1$

6. Let $(1+x+x^2)^n = a_0 + a_1x + a_2x^2 ..... + a_{2n}x^{2n}$, where n is odd integer, if

$S_1 = a_0 + a_4 + a_8 + ...$

$S_2 = a_1 + a_5 + a_9 + ...$

$S_3 = a_2 + a_6 + a_{10} + ...$

$S_4 = a_3 + a_7 + a_{11} + ...$

then

(a) $S_1 = S_3$ (b) $S_2 = S_4$

(c) $S_2 + S_4 = 0$ (d) either $S_1 = S_2 = S_3$ or $S_1 = S_3 = S_4$

7. $\cos\alpha$ is a root of the equation $169x^2 - 26x - 35 = 0$, $-1 < x < 0$, then $\sin 2\alpha$ is

(a) $\frac{144}{169}$ (b) $-\frac{144}{169}$ (c) $\frac{120}{169}$ (d) $-\frac{120}{169}$

8. A solution (x, y) of the system of equations $x - y = \frac{1}{3}$ and $\cos^2(\pi x) - \sin^2(\pi y) = \frac{1}{2}$ is given by

(a) $\left(\frac{7}{6}, \frac{5}{6}\right)$ (b) $\left(\frac{2}{3}, \frac{1}{3}\right)$ (c) $\left(-\frac{5}{6}, -\frac{7}{6}\right)$ (d) $\left(\frac{13}{6}, \frac{11}{6}\right)$

SECTION – III
Reasoning Type

This section contains 4 reasoning type questions. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True; **Statement-2** is a correct explanation for **Statement-1**.

(B) **Statement-1** is True, **Statement-2** is True; **Statement-2** is not a correct explanation for **Statement-1**.

(C) **Statement-1** is True, **Statement-2** is False.

(D) **Statement-1** is False, **Statement-2** is True.

9. **Statement-1 :** For a conic section if its two foci are (2, 3), (10, 12) and its directrix is $8x + 9y - 50 = 0$, then it must be hyperbola.

Statement-2 : For a hyperbola, line joining the foci is perpendicular to directrix.

(a) A (b) B (c) C (d) D

10. **Statement-1 :** $\lim_{x\to 0}\left(\left(\frac{1+2x}{1+3x}\right)^{\frac{1}{x^2}} e^{\frac{1}{x}}\right) = \lim_{x\to 0} e^{\left\{\frac{1}{x^2}\left(\frac{1+2x}{1+3x}-1\right)+\frac{1}{x}\right\}} = e^{\lim_{x\to 0}\frac{1}{(1+3x)x}+\frac{1}{x}} = e^{\lim_{x\to 0}\frac{3}{1+3x}} = e^3$.

Statement-2 : If $\lim_{x\to a} f(x) = l_1$ and $\lim_{x\to a} g(x) = l_2$, then $\lim_{x\to a}(f(x)g(x)) = l_1.l_2$

(where l_1 and l_2 are finite).

(a) A (b) B (c) C (d) D

11. **Statement 1 :** $\lim\limits_{t\to 0}\int_0^{2\pi}\frac{|\sin(x+t)-\sin x|}{|t|}dx$ is equals to 4.

Statement 2 : $f'(x)=\lim\limits_{h\to 0}\frac{f(x+h)-f(x)}{h}$.

(a) A (b) B (c) C (d) D

12. **Statement-1 :** Let $f:R\to R,\ f(x)$ be a integrable function then

$$\int_0^1 f^2(x)\,dx \geq \left(\int_0^1 f(x)\,dx\right)^2.$$

Statement-2 : $\sum\limits_{r=1}^{n}\left(\frac{a_r^2}{n}\right)\geq\left(\sum\limits_{r=1}^{n}\frac{a_r}{n}\right)^2$.

(a) A (b) B (c) C (d) D

SECTION–IV
Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Passage-I

Consider an isosceles triangle *ABC* with base *BC* = 8 units and $\angle B = \angle C$. A line cuts the base externally at *D* such that it make angle θ with base and bisects the side which is nearer to *D*.

13. If $\angle B = 45°$, then the area of quadrilateral which the straight line cuts from the given triangle is

(a) $\frac{3+\tan\theta}{1+\tan\theta}$ (b) $\frac{3+5\tan\theta}{1+\tan\theta}$ (c) $\frac{2(3+5\tan\theta)}{1+\tan\theta}$ (d) $\frac{4(3+5\tan\theta)}{1+\tan\theta}$

14. If $\angle B = 45°$, then length of portion of straight line inside the triangle may lie in the range

(a) (2, 4) (b) $(2\sqrt{2}, 4)$ (c) $(\sqrt{2}, 2)$ (d) none of these

15. If the vertices of triangle *ABC* lie on the curve $x^3 + y^3 + 3xy = 1$, then number of such possible triangles is

(a) 1 (b) 2 (c) 3 (d) 4

Passage-II

$f(x) = a_0 + a_1 \cos x + a_2 \cos 2x + a_3 \cos 3x + \ldots a_n \cos nx$

$g(x) = b_1 \sin x + b_2 \sin 2x + b_3 \sin 3x + \ldots b_n \sin nx$

16. If $b_r = \dfrac{1}{r+1}$, then $\lim\limits_{n\to\infty} \int_0^{\pi} g(x)\,dx$ is

(a) $\frac{1}{2}\ln\frac{1}{2}$ (b) $\frac{1}{2}\ln 2$ (c) $\ln 2$ (d) $2\ln 2$

17. $\sum\limits_{k=0}^{n} \int_0^{2\pi} f(x).(\cos kx)\,dx$ is

(a) $\pi f(0)$ (b) $\pi f(1)$ (c) $\pi f(-1)$ (d) 0

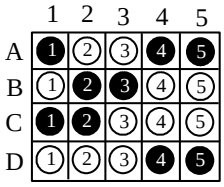
18. $\sum\limits_{k=0}^{n} \int_0^{2\pi} g(x).(\cos kx)\,dx$ is

(a) $\pi f(0)$ (b) $\pi g(1)$ (c) $\pi g(-1)$ (d) 0

SECTION–V
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



1.

Column I (Function)		Column II (Property)
(A)	$f: N \to I \quad f(x) = (x+1)^2 + x - [\sqrt{(x+1)^2 + (x+1)}]^2$, then $f(x)$ will be (where [.] is greatest integer function)	**1.** one-one function
(B)	Let $f(x) = \frac{\log x}{x}$, and solutions of $f(x) = k$ be denoted by $g(k)$ {where $k < 0$}, $g: (-\infty, 0) \to (0, 1)$, then $y = g(x)$ will be	**2.** many one function
(C)	Let $f: R \to R$ be a continuous function such that $\lvert f(x) - f(y) \rvert \geq \lvert x - y \rvert$ for all $x, y \in R$, then $f(x)$ will be	**3.** onto function
(D)	$f: R \sim \{0\} \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ such that $f(x) = \int_0^x \frac{1}{1+t^2} dt + \int_0^{1/x} \frac{1}{1+t^2} dt$ ($x \neq 0$), now $f(x)$ will be	**4.** into function **5.** neither one-one nor onto

2.

Column I	**Column II**
(A) Let $\lvert z_1\rvert=\lvert z_2\rvert=\lvert z_3\rvert=1$, $\frac{z_1^2}{z_2z_3}+\frac{z_2^2}{z_1z_3}+\frac{z_3^2}{z_1z_2}+1=0$ and $\lvert z_1+z_2+z_3\rvert=k$, then k can be	**1.** 1
(B) Let $\lvert z\rvert=1$ and $a\le\lvert 1-z\rvert+\lvert 1+z^2\rvert\le b$, and k be a integer such that $k\in[a\ b]$, then k can be	**2.** 2
(C) Let $\lvert z\rvert=1,\lvert 1-z\rvert=k\sin\frac{\theta}{2}$, (where $\theta=\arg(z)$), then (k+2) can be	**3.** 3
(D) Let $f(x)=x^4+ax^3+bx^2+cx+d$ (where a, b, c, d are real coefficients) and $f(x)=0$ has real roots. If $\lvert f(i)\rvert=1$ and $a=b=c=d=k$, then k can be	**4.** 4 **5.** 0

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PAPER-II
PHYSICS
SECTION – I
Single Correct Choice Type

1. Slide callipers can measure the external and internal diameters as well as the depth of a hollow cylinder. This is sufficient to find its volume and hence density, since mass is known.

$\therefore$ **(a)**

2. $I = ne\pi r^2 v$ [here n = no. of protons per unit volume and v is velocity of proton]

$$\varepsilon_0 (2\pi r v).\ E = I$$

$$\therefore \quad E = \frac{I}{2\pi r \varepsilon_0 v}$$

$$\frac{1}{2} m v^2 = \text{eV}$$

$$v = \sqrt{\frac{2eV}{m}}$$

$$E = \frac{I}{2\pi r \varepsilon_0} \sqrt{\frac{m}{2\,\text{eV}}}$$

$\therefore$ **(a)**

3. $\sqrt{\frac{3RT}{M}} =$ velocity of sound

$$\sqrt{\frac{3 \times R \times 273}{M}} = 330$$

$$\sqrt{\frac{3R}{M}} = \frac{330}{\sqrt{273}} \approx 20$$

Variation of temperature with height $T = T_1 - \left(\frac{T_1 - T_2}{y_2}\right) y$

On differentiation $1 = -\frac{(T_1 - T_2)}{y_2} \cdot \frac{dy}{dT}$

$$V = \sqrt{\frac{3RT}{M}} = \frac{dy}{dt} \cdot \frac{dT}{dT} = \frac{dy}{dT} \cdot \frac{dT}{dt}$$

$$-\frac{20(T_1 - T_2)}{y_2} \int_0^t dt = \int_{T_1}^{T_2} \frac{dT}{\sqrt{T}}$$

t = 30 sec

$\therefore$ **(a)**

4. $dQ = dU + dW$

$dW = Pdv = 10^5\left(0.5\times10^{-3}\right) = 50\,\text{J}$

$dU = 500 - 50 = 450$

$450 = a\dfrac{dT}{T} = a\dfrac{\left(T - T_0\right)}{T}$

$\dfrac{450}{a} = 1 - \dfrac{T_0}{T}$

$\dfrac{T_0}{T} = 1 - \dfrac{450}{3000} = \dfrac{2550}{3000}$

$T = 1.17\ T_0$

$\therefore$ **(b)**

SECTION – II
Multiple Correct Choice Type

5. Total energy $= \dfrac{1}{2}\dfrac{q_0^2}{C}$

From LC circuits $L\dfrac{dI}{dt} + \dfrac{q}{c} = 0$

The solution of above equation is $q = q_0\cos\omega t\ \left(\omega = \dfrac{1}{\sqrt{LC}}\right)$

At an instant $t = \dfrac{T}{8}$

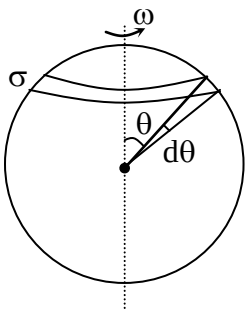
$q = q_0\cos\dfrac{2\pi}{T}\dfrac{T}{8} = q_0/\sqrt{2}$

$U_e = \dfrac{q^2}{2C} = \dfrac{{q_0}^2}{4C}$, $U_B = 50\%$ of total energy

Ratio is 1

$\therefore$ **(a)(b) and(d)**

6. Charge on differential strip $dq = \left(2\pi R\sin\theta\right)\left(Rd\theta\right).\sigma$



$dI = \dfrac{\omega dq}{2\pi} = \omega\sigma R^2\sin\theta d\theta$

Magnetic field $= \dfrac{\mu_0 dI\left(R^2\sin^2\theta\right)}{2R^3} = \dfrac{\mu_0\omega rR\sin^3\theta\, d\theta}{2}$

$B = \dfrac{\mu_0\omega\sigma R}{2}\int_0^{\pi}\sin^3\theta d\theta = \dfrac{2}{3}\mu_0\omega\sigma R$

Magnetic moment due to it

$\mu = \int dI\ \pi r^2 = 2\int_0^{\pi/2}\left(\omega\sigma R^2\sin\theta\right)\pi R^2\sin^2\theta d\theta = \dfrac{4}{3}\pi\sigma R^4\omega$

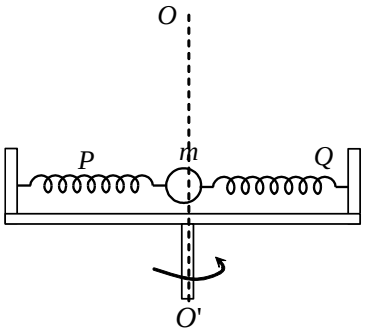
$\therefore$ **(a) and (b)**

7. $m\frac{d^2x}{dt^2} = -kx + m\omega^2 x$

$\frac{d^2x}{dt^2} = \left(-\frac{k}{m} + \omega^2\right)x$

Here $\omega_0 = \sqrt{\frac{k}{m} - \omega^2}$

$T = \frac{2\pi}{\sqrt{\frac{k}{m} - \omega^2}} = 0.70\text{sec}$



The oscillation will stop if $T = \infty$

It is possible when $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{28.8}{0.2}} = 12$ rad/sec

∴ **(a) and (c)**

8. $\tau = T_1 = (\mu R_1 + \mu R_2)r$

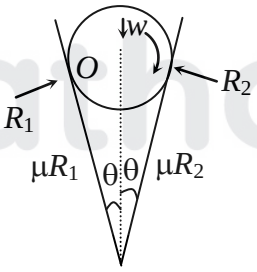
$w = (R_1 + R_2)\sin\theta + \mu(R_2 - R_1)\cos\theta$

In horizontal direction

$(R_2 - R_1)\cos\theta = \mu(R_1 + R_2)\sin\theta$

$w = (R_1 + R_2)(1 + \mu^2)\sin\theta$

$T_1 = \frac{\mu wr}{(1+\mu^2)\sin\theta}$



When sphere leaves contact with O

$R_1 = 0$, $T = \mu R_2 r$

$w = R_2(\sin\theta + \mu\cos\theta)$

$R_2 = \frac{w}{(\sin\theta + \mu\cos\theta)}$

In horizontal direction

$R_2\cos\theta = \mu R_2\sin\theta$

$\mu = \frac{\cos\theta}{\sin\theta}; T = \frac{(\cot\theta).wr}{\sin\theta + \frac{\cos^2\theta}{\sin\theta}} = wr\cos\theta$

∴ **(a), (b) and(c)**

SECTION – III
Reasoning Type

9. Momentum of the system of particles depends only on $\vec{F}_{ext}$ though momentum and kinetic energy of individual particle may change due to internal forces.

$\therefore$ **(d)**

10. $|\vec{R}_1| = \sqrt{A^2 + B^2 + 2AB\cos\phi}$

Now, $|\vec{R}_1| = |\vec{A}| = |\vec{B}| = 1$

So $\phi = 120^0$

$|\vec{R}_2| = \sqrt{A^2 + B^2 - 2AB\cos\phi}$

Now, $|\vec{A}| = |\vec{B}| = 1$

So $|\vec{R}_2| = \sqrt{3}$

$\therefore$ **(b)**

11. If images are formed by TIR, transmission do not take place. So total light is reflected back to form brighter image.

$\therefore$ **(a)**

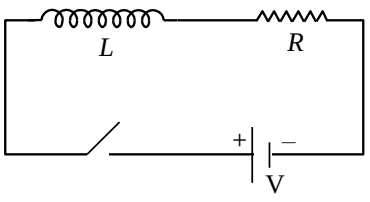
12. Kinetic energy = $h\nu$ - ϕ so K.E. $\propto \nu$

$$\text{K.E'} = \frac{h\nu}{2} - \phi \Rightarrow \text{K.E} \neq 2\,\text{K.E'}$$

$\therefore$ **(d)**

SECTION–IV
Comprehension Type
Passage-I

The rate of increase of energy is the difference between output power and power dissipated in resistor



$$\frac{d(E_{meg})}{dT} = VI - RI^2 = -R\left[I - \frac{V}{2R}\right]^2 + \frac{V^2}{4R} \leq \frac{V^2}{4R}$$

It is clear that rate of increase in maximum when $I = V/2R = 10/2\times 5$ = 1 amp

Only at $t = \infty$

$$I = \frac{V}{R} = \frac{10}{5} = 2 \text{ amp}$$

The power dissipation in resistor is

$$P = I^2 R = \frac{V^2}{R} = \left(1 - e^{-R/L.t}\right)^2$$

Fastest rate of change of energy becomes for both the inductor and resistor, when I^2 is changing most rapidly and this vacuums at $I = V/2R$

Put $I = V/2R$, we get $e^{-(R/L)t} = \frac{1}{2}$

$$t = 0.69\left(\frac{L}{R}\right) = 0.69 \times \left(\frac{500}{5}\right) = 69 \sec$$

13. **(a)**

14. **(b)**

15. **(a)**

Passage-II

$$dR = \frac{1}{\sigma(x)}\frac{dx}{A} = \frac{\sqrt{x}dx}{Al\sigma_0}$$

$$R = \int_0^l dR = \frac{1}{Al\sigma_0}\int_0^l \sqrt{x}dx = \frac{2\sqrt{l}}{3A\sigma_0}$$

$$I = \frac{v_0}{R} = \frac{3Av_0\sigma_0}{2\sqrt{l}}$$

$$J = \frac{I}{A} = \frac{3\sigma_0 v_0}{2\sqrt{l}}$$

$$E(x) = \frac{T}{\sigma(x)} = \frac{T}{\sigma_0 \frac{l}{\sqrt{x}}} = \frac{3v_0\sqrt{x}}{2(l)^{3/2}}$$

16. **(d)**

17. **(a)**

18. **(a)**

SECTION–V
Matrix–Match Type

1. **(A)** $\beta = \dfrac{\lambda D}{d}$

So on increasing D, β will increases

(B) $\beta = \dfrac{\lambda D}{d}$

So on increasing d, β will decrease

(C) $\lambda_w = \dfrac{\lambda air}{\mu_W}$ so $\beta_W = \dfrac{\lambda_{air} D}{\mu_W d}$ so, $\beta_W < \beta_{air}$

(D) Due to introduction of slab path difference at y = 0 is not zero

So all the fringes along with central fringe shift by same amount.

$\therefore$ **(A) – 1, 5; (B) – 2, 5; (C) – 2, 5; (D) – 3, 4**

2. **(A)** $(V_{AB})_v = 40\,\text{m/s} \;\therefore\; (S_{AB})_v = (40\times 2) = 80\,\text{m}$

$(V_{AB})_H = 0\,\text{m/s} \;\therefore\; (S_{AB})_H = 0\times 2 = 0\,\text{m}$

(B) $(V_{AB})_v = 40\,\text{m/s} \;\therefore\; (S_{AB})_v = (40\times 2) = 80\,\text{m}$

$(V_{AB})_H = 0\,\text{m/s} \;\therefore\; (S_{AB})_H = 0\times 2 = 0\,\text{m}$

(C) $(V_{AB})_v = 16\,\text{m/s}$, $(V_{AB})_H = 8\,\text{m/s}$

$\therefore\; V_{AB} = 8\sqrt{5}\,\text{m/s}$

$S_{AB} = 8\sqrt{5}\times 2 = 16\sqrt{5}\,\text{m}$

(D) $(V_{AB})_v = 24\,\text{m/s} \;\therefore\; (V_{AB})_H = 32\,\text{m/s}$

$\therefore\; V_{AB} = 40\,\text{m/s}$, $S_{AB} = 40\times 2 = 80\,\text{m}$

$(S_{AB})_H = 2\times 32 = 64\,\text{m}$

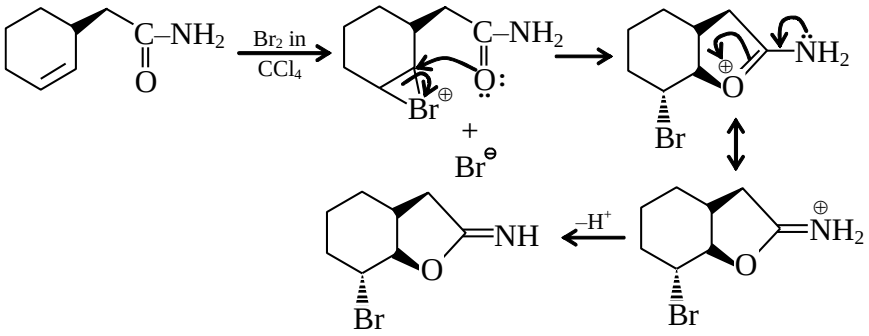
$\therefore$ **(A) – 2, 4, 5; (B) – 2, 4, 5; (C) – 1, 3; (D) – 2, 4**

CHEMISTRY

SECTION – I

Single Correct Choice Type

1. (c)



2. (d)

$$HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$$

Initially	c	0	0
At equilib.	$c(1-\alpha)$	$c\alpha$	$c\alpha$

$$K_a = \frac{(c\alpha)^2}{c(1-\alpha)} = \frac{(0.1\,c)^2}{0.9\,c} = \frac{0.01\,c}{0.9}$$

$$K_a = \frac{0.1\,c}{9} \quad ; \quad c = \frac{9\,K_a}{0.1} = \mathbf{90\ K_a.}$$

3. (c)

$$A \underset{k_r}{\overset{k_f}{\rightleftharpoons}} B \quad \text{(Mutarotates)}$$

t = 0	a	0
t = t	a – *x*	*x*
t = ∞	a – y	y

Rate of forward reaction, $\frac{dx}{dt} = k_f(a - x)$

Rate of reverse reaction, $\frac{-dx}{dt} = k_r x$

Observed rate of reaction, $\frac{dx}{dt} = k_f(a - x) - k_r x$

$$= k_f a - (k_f + k_r)x \qquad \ldots(1)$$

Let, 'y' be the value of *x* at equilibrium.

At equilibrium, rate of forward reaction = rate of reverse reaction.

$$k_f(a - y) = k_r y$$

$$k_f a - k_f y = k_r y$$

$$a = \frac{(k_f + k_r)}{k_f}.y$$

Substituting value of 'a' in equation (1), gives

$$\frac{dx}{dt} = k_f \frac{(k_f + k_r)}{k_f} y - (k_f + k_r)x = (k_f + k_r)(y - x)$$

$$\int_0^x \frac{dx}{(y - x)} = (k_f + k_r)\int_0^t dt$$

$$\left[-\ln(y - x)\right]_0^x = (k_f + k_r)\, t$$

$$-\ln (y - x) + \ln y = (k_f + k_r)\, t$$

$$(k_f + k_r) = \frac{1}{t} \ln \frac{y}{(y - x)}$$

Let r_A and r_B be the optical rotations for 1 mole of A & B respectively.

$r_o = ar_A$

$r_t = (a - x)r_A + xr_B = ar_A + x(r_B - r_A) = r_o + x(r_B - r_A)$

$$x = \frac{(r_t - r_o)}{(r_B - r_A)}$$

$r_\infty = (a - y)r_A + yr_B = ar_A + y(r_B - r_A) = r_o + y(r_B - r_A)$

$$y = \frac{(r_\infty - r_o)}{(r_B - r_A)},\ (y - x) = \frac{(r_\infty - r_t)}{(r_B - r_A)}$$

$$\therefore \quad \frac{y}{y - x} = \frac{(r_\infty - r_o)}{(r_\infty - r_t)} = \frac{30 - 190}{30 - 150} = \frac{-160}{-120} = \frac{4}{3}.$$

$$\therefore \quad (k_f + k_r) = \frac{2.303}{180} \log \frac{4}{3} = \frac{2.303}{180} \times 0.123$$

$(k_f + k_r) = 1.57 \times 10^{-3}$ min^{-1}.

4. **(c)** Bonding in I_3^- arises from singly occupied $5P_z$ orbitals on two iodine atoms and a full $5P_z$ orbital on I^-, then there are three atomic orbitals involved. Thus, 3 molecular orbitals are formed, one bonding, one non-bonding and 1 antibonding. There are 4 e^-'s, 2 occupy bonding MO and 2 occupy the non-bonding MO. This gives one bond, spread over two positions i.e. the bond order of I–I bond in I_3^- is 0.5.

5. **(d)** $[Fe(CO)_5]$ is trigonal bipyramidal in shape. $[Ni(DMG)_2]$ is a rosy-red precipitate having square planar shape. $[Ni(CN)_4]^{2-}$ is colourless and diamagnetic with $\mu = 0$. $[Fe(CN)_5NO]^{2-}$ has strong field ligands allowing d^2sp^3 hybridization.

6. **(c)** $$R{-}OR' + H{-}I \longrightarrow R{-}OH + R'{-}I \xrightarrow[\text{solution}]{\text{Alcoholic } AgNO_3} AgI\downarrow + R'^{+}NO_3^-$$

Ether with one mole of methoxy group reacts with one mole of HI to form one mole of CH_3I, which can react with alcoholic $AgNO_3$ to give 1 mole of AgI precipitate. Similarly, 1 mole of ether with 'n' moles of methoxy group reacts with 'n' moles of HI to give 'n' moles of AgI precipitate.

Moles of organic compound = $\frac{0.2063}{168}$

Moles of AgI produced = $n \times \frac{0.2063}{168} = \frac{0.8658}{235}$

$n = 3.$

7. **(c)** SiC does not get hydrolysed as it is a covalent carbide. Na_2C_2 on hydrolysis gives C_2H_2 (So, it's an acetylide) and Mg_2C_3 on hydrolysis gives $CH_3C{\equiv}CH$ (So, it's a propynide). Mn_3C on hydrolysis liberates CH_4 and H_2, so it is a methanide.

$$Mn_3C + 6H_2O \longrightarrow 3Mn(OH)_2 + CH_4\uparrow + H_2\uparrow$$

8. **(d)** Let total number of moles of Ni are *x*, out of which 96% are in +2 state and 4% are in +3 state.

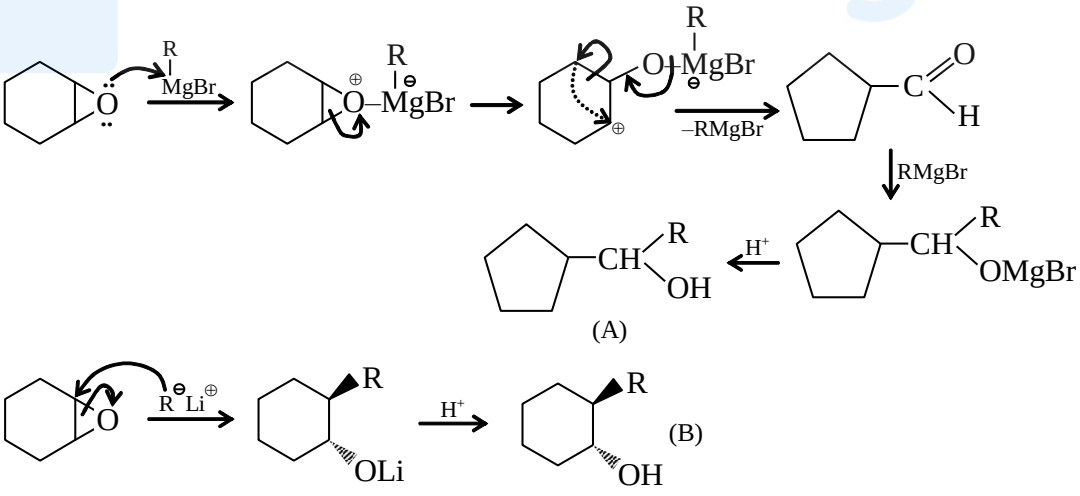
Maintaining the electroneutrality,

$$\left[x \times \frac{96}{100}(+2)\right] + \left[x \times \frac{4}{100}(+3)\right] = 2$$

$$x = \frac{200}{204} = \frac{50}{51} = \mathbf{0.98}$$

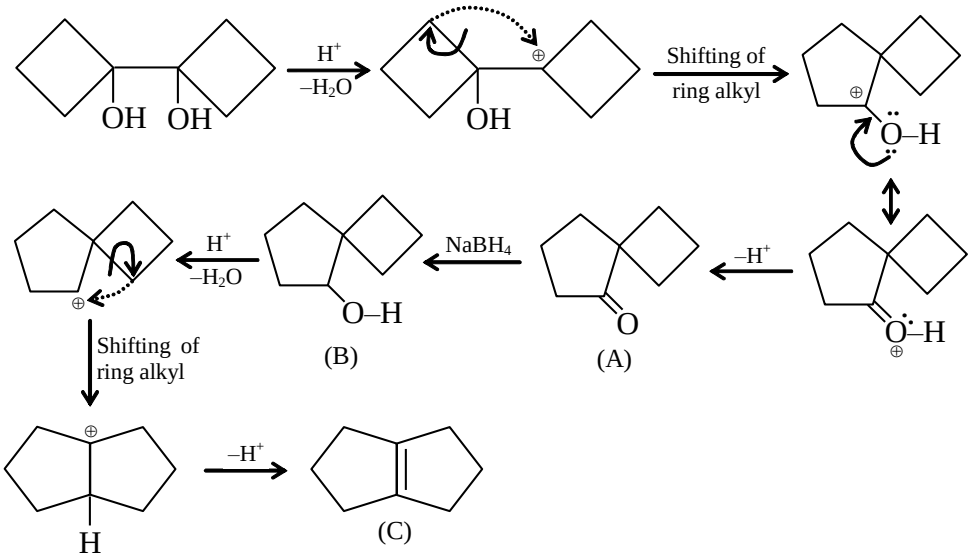
SECTION – II
Multiple Correct Choice Type

9. **(a, b)**



10. **(d)** ΔG at equilibrium is zero and thus cannot be related to K_p or K_c. It is $\Delta G°$, that is related to K_p or K_c, provided K_p and K_c both exist for a given equilibrium. For equilibrium involving solution species, K_p can not exist.

11. (b, c)



12. (a, b & d)

Let 'n' moles of A and B are present in solution. So, the moles of A in the solution is $\frac{n}{3}$.

Total number of moles in the distillate = $\frac{n}{4}$

Number of moles of A in the distillate = $\frac{n}{4}\times\frac{3}{4} = \mathbf{\frac{3n}{16}}$

Total pressure of solution, $P_T = P_A^o X_A + P_B^o X_B$

$$600 = 300 \times \frac{1}{3} + P_B^o \cdot \frac{2}{3}$$

$$\frac{3\times 500}{2} = P_B^o$$

$$P_B^o = \frac{1500}{2} = \mathbf{750\ mm\ Hg}.$$

Total number of moles in the residue = $\frac{3n}{4}$

Moles of A in the residue = $\frac{n}{3}-\frac{3n}{16} = \frac{7n}{48}$

Mole fraction of A in residue = $\frac{\frac{7n}{48}}{\frac{3n}{4}} = \frac{28}{48\times 3} = \frac{7}{36}$

Mole fraction of B in residue = $1 - \frac{7}{36} = \mathbf{\frac{29}{36}}$

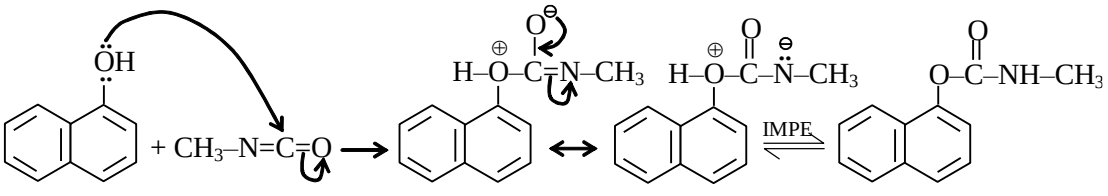
Total pressure of the residue $= \left(300\times\frac{7}{36}\right)+\left(\frac{1500}{2}\times\frac{29}{36}\right)$

$= 58.33 + 604.17 = \mathbf{662.5\ mm\ Hg}$.

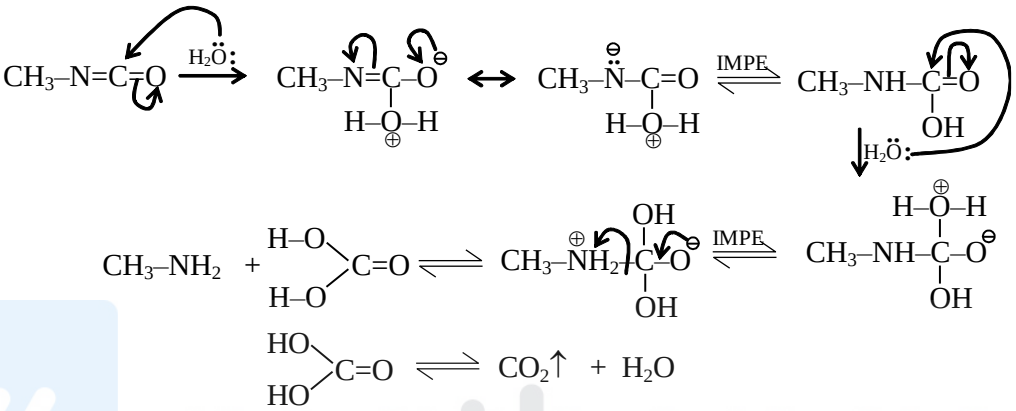
SECTION–III
Comprehension Type

Passage–I

13. **(b)**



14. **(c)**



15. **(d)** Due to intermolecular H-bonding between $CH_3–NH_2$ and H_2O, energy will be released, thus raising the temperature of the system.

Passage–II

16. **(b)** $f = \dfrac{99.9/0.1}{0.1/99.9} = (999)^2$

$$f' = \frac{r_{^{235}UF_6} / r_{^{238}UF_6}}{n_{A_o} / n_{B_o}} = \sqrt{\frac{M_{^{238}UF_6}}{M_{^{235}UF_6}}} = \sqrt{\frac{352}{349}}$$

$(f')^m = f$ where 'm' is the number of effusion steps needed to get desired concentration ratio.

$$\left(\frac{352}{349}\right)^{\frac{m}{2}} = (999)^2$$

$$\frac{m}{2}[\log 352 - \log 349] = 2 \log 999 = 2 \times 3$$

$$\frac{m}{2}[2.546 - 2.542] = 2 \times 3$$

$$m(0.004) = 12$$

$$m = \frac{12}{4\times10^{-3}} = 3 \times 10^3 = \textbf{3000 steps}.$$

17. **(c)** $f = \frac{64/1}{2/1} = 32 = 2^5$

$f' = \frac{r_{H_2}/r_{He}}{n_{H_2}/n_{He}} = \sqrt{\frac{M_{He}}{M_{H_2}}} = \sqrt{\frac{4}{2}} = \sqrt{2}$

$(f')^m = f$

$(2)^{m/2} = 2^5$

$\frac{m}{2} = 5$

$\therefore$ m = **10**

18. **(b)** $f = \frac{n_{H_2}/n_{D_2}}{10/90}$

$f' = \frac{r_{H_2}/r_{D_2}}{10/90} = \sqrt{\frac{M_{D_2}}{M_{H_2}}} = \sqrt{\frac{4}{2}} = \sqrt{2}$

$(f')^m = f$

$(2)^{m/2} = \frac{n_{H_2}/n_{D_2}}{10/90}$; $(2)^{18/2} \times \frac{10}{90} = \frac{n_{H_2}}{n_{D_2}}$

$2^9 \times \frac{1}{9} = \frac{n_{H_2}}{n_{D_2}}$; $\frac{512}{9} = 56.88 = \frac{n_{H_2}}{n_{D_2}}$

$\therefore$ % of H_2 in final mixture = $\frac{56.88}{57.88} \times 100 \simeq$ **98%**.

SECTION–IV
Matrix–Match Type

1. **(A) – 1, 2, 3, 4, 5 ; (B) – 3, 4, 5 ; (C) – 3, 4, 5 ; (D) – 2, 3**

2. **(A) – 1, 2, 3, 4, 5 ; (B) – 1, 2 ; (C) – 1, 2, 3, 4, 5 ; (D) – 2, 3, 5**

MATHEMATICS

SECTION – I

Single Correct Choice Type

1. (b) Option (b) is correct because in this case $a > 0,\ b < 0$ (let $b = -c$)

$$\Rightarrow \quad -cx^2 + ay^2 = -ac \quad \Rightarrow \quad \frac{x^2}{a} - \frac{y^2}{c} = 1$$

2. (a) $f : R \to R,\ f(x) = \dfrac{x^2 + bx + 1}{x^2 + 2x + b}$, Range must be of the form $\left[\dfrac{1}{a}, a\right]$

So numerator and denominator never be zero.

$\Rightarrow \ D < 0 \ \Rightarrow \ 4 - 4b < 0 \ \Rightarrow \ b > 1$ (for denominator)

$\Rightarrow \ b^2 < 4 \ \Rightarrow \ -2 < b < 2$ (for numerator)

$$y = \frac{x^2 + bx + 1}{x^2 + 2x + b} \quad \Rightarrow \quad (y-1)x^2 + (2y - b)x + by - 1 = 0$$

If we make its discriminant zero, then its equation has root of form $a, \dfrac{1}{a}$.

$$\Rightarrow \quad 4(1-b)y^2 + 4y + b^2 - 4 = 0 \ \Rightarrow \quad \frac{b^2 - 4}{4(1-b)} = 1 \quad \Rightarrow \quad b = -2 \pm 2\sqrt{3}$$

3. (b) $(\cos y - \sin y)\dfrac{dy}{dx} + (\cos y + \sin y)e^{-x} = e^{e^{-x}}$

$\cos y + \sin y = u$

$$\Rightarrow \quad \frac{du}{dx} + e^{-x}u = e^{e^{-x}} \quad \Rightarrow \quad ue^{-e^{-x}} = x$$

$$\Rightarrow \quad e^{-e^{-t}} = t \qquad \text{(putting } (t, 0)\text{)}$$

$$\Rightarrow \quad t.e^{e^{-t}} = 1$$

4. (c) Multiply by y, z and x in rows 1, 2 and 3 respectively and then take common y, z and x from column 1, 2 and 3 respectively, then

$$\begin{vmatrix} y^3+1 & y^3 & y^3 \\ z^3 & z^3+1 & z^3 \\ x^3 & x^3 & x^3+1 \end{vmatrix} = 11$$

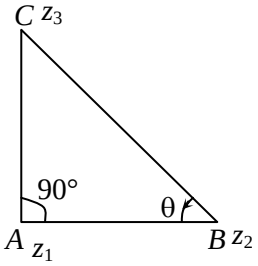
$$\Rightarrow \begin{vmatrix} 1 & 0 & y^3 \\ -1 & 1 & z^3 \\ 0 & -1 & x^3+1 \end{vmatrix} = 11 \qquad \left(C_1 \to C_1 - C_2 \text{ and } C_2 \to C_2 - C_3\right)$$

$$\Rightarrow \ 1\left(x^3 + 1 + z^3\right) + y^3(1) = 11 \ \Rightarrow \ x^3 + y^2 + z^3 = 10$$

So solution are (1, 1, 2), (1, 2, 1) or (2, 1, 1)

SECTION – II
Multiple Correct Choice Type

5. **(a), (b)**



$$\frac{W_2}{|W_2|} = \frac{W_1}{|W_1|}e^{-i\theta}$$

$$\Rightarrow \quad \frac{W_2}{W_1} = \frac{|W_2|}{|W_1|}e^{-i\theta} = 1 - i\tan\theta$$

$$W_1\overline{W_2} = \frac{W_1}{W_2}|W_2|^2 = |W_1||W_2|e^{i\theta} = \left(|W_1|^2 + i|W_1||W_2|\sin\theta\right)$$

6. **(a), (d)**

$$(1+x+x^2)^n = a_0 + a_1x + a_2x^2 + \ldots + a_{2n}x^{2n} \qquad \ldots(1)$$

$$\Rightarrow \quad a_0 + a_2 + a_4 \ldots = \frac{3^n+1}{2}$$

$$a_1 + a_3 + a_5 \ldots = \frac{3^n-1}{2}$$

Put $x = i$ in equation (1) where $i = \sqrt{-1}$

If $n = 4m+1$

$\Rightarrow \quad a_1 - a_3 + a_5 \ldots = 1$

$a_0 - a_2 + a_4 \ldots = 0$

So, $a_1 + a_5 + a_9 \ldots = \frac{3^n+1}{4} = S_2$

$$a_3 + a_7 + a_{11} \ldots = \frac{3^n-3}{4} = S_4$$

Similarly if $n = 4m+3$

Then, $S_2 = \frac{3^n-3}{4}$ and $S_4 = \frac{3^n+1}{4}$

7. **(c), (d)**

$$(13x-1)^2 = 36$$

$$13x - 1 = \pm 6$$

$$x = \frac{7}{13}$$

$$x = -\frac{5}{13}$$

$$\Rightarrow \quad \cos\alpha = -\frac{5}{13}; \quad \sin\alpha = \pm\frac{12}{13}$$

$$\sin 2\alpha = \pm\frac{120}{169}$$

8. **(a), (c), (d)**

$$\cos^2(\pi x) - \sin^2(\pi y) = \frac{1}{2}$$

$$\Rightarrow \quad \cos\{\pi(x+y)\}\cos\{\pi(x-y)\} = \frac{1}{2}$$

$$\Rightarrow \quad \cos\{\pi(x+y)\}\cos\left(\frac{\pi}{3}\right) = \frac{1}{2} \qquad \left(\because x - y = \frac{1}{3}\right)$$

$$\Rightarrow \quad \cos\pi\{x+y\} = 1 \;\Rightarrow\; \pi(x+y) = 2n\pi \;\Rightarrow\; x + y = 2n,\ n \in I$$

Now, $x + y = 2n$ and $x - y = \frac{1}{3}$

$$\Rightarrow \quad x = n + \frac{1}{6},\ y = n - \frac{1}{6}$$

$$\Rightarrow \quad (x,\ y) = \left(n + \frac{1}{6},\ n - \frac{1}{6}\right)$$

(a), (c), (d) correspond $n = 1, -1, 2$ respectively.

SECTION – III
Reasoning Type

9. **(b)** As both foci lies on opposite side of directrix the curve must be hyperbola and statement 2 is true but does not explain the first one.

10. **(d)** $\lim\limits_{x\to 0}\left(\frac{1+2x}{1+3x}\right)^{\frac{1}{x^2}} e^{\frac{1}{x}} = \lim\limits_{x\to 0} e^{\frac{\ln(1+2x)-\ln(1+3x)+x}{x^2}}$ (using L' hospital rule)

$= e^{5/2}$

11. **(a)** $\lim\limits_{t\to 0}\int_0^{2\pi}\left|\frac{\sin(x+t)-\sin x}{t}\right| dx = \int_0^{2\pi} |\cos x|\, dx = 4$

12. **(a)** As we know that $\left(a_1^2 + a_2^2 ..... + a_n^2\right)\left(b_1^2 + b_2^2 ..... + b_n^2\right) \geq \left(a_1 b_1 + a_2 b_2 + ... + a_n b_n\right)^2$

Now, put $b_1 = b_2 = b_3 ..... b_n = 1$

Now let $a_r = f\left(\frac{r}{n}\right)$

$$\Rightarrow \quad \sum_{r=1}^{n} f^2\left(\frac{r}{n}\right)\frac{1}{n} \geq \left(\sum_{r=1}^{n}\frac{f(r/n)}{n}\right)^2 \;\Rightarrow\; \lim_{n\to\infty}\sum_{r=1}^{n} f^2\left(\frac{r}{n}\right)\frac{1}{n} \geq \left(\lim_{n\to\infty}\sum_{r=1}^{n}\frac{f(r/n)}{n}\right)^2$$

$$\Rightarrow \quad \int_0^1 f^2(x)dx \geq \left(\int_0^1 f(x)dx\right)^2$$

SECTION–IV
Comprehension Type

Passage-I

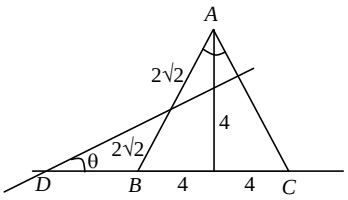
Sol.:

13. (d) Area of quadrilateral will be

$$= 16 - \frac{1}{2} 2\sqrt{2}.2\sqrt{2} \tan(45 - \theta)$$

$$= 16 - 4\left(\frac{1 - \tan\theta}{1 + \tan\theta}\right)$$

$$= \frac{4(4 + 4\tan\theta - 1 + \tan\theta)}{1 + \tan\theta} = 4\left(\frac{3 + 5\tan\theta}{1 + \tan\theta}\right)$$



14. (b) Length of portion of straight line inside the triangle = $2\sqrt{2}\sec\left(\frac{\pi}{4} - \theta\right)$.

15. (c) $x^3 + y^3 + (-1)^3 - 3(-1)(x)(y) = 0$

$\Rightarrow (x + y - 1)[(x - y)^2 + (x + 1)^2 + (y + 1)^2] = 0$

$\Rightarrow$ one point (–1, –1) and two points satisfies the equation $x + y = 1$.

So, three isosceles triangle are possible such that one side length 8 and two angles are equal.

Passage-II

16. (d) $\int_0^{\pi} g(x)\,dx =$ Imaginary part of $\sum_{r=1}^{n} \int_0^{\pi} b_r e^{irx}\,dx$

$$= \sum_{r=1}^{n} \frac{2b_r}{r} \qquad (r \text{ is odd integer})$$

$$\lim_{n\to\infty} \int_0^{\pi} g(x)\,dx = 2\left(\frac{b_1}{1} + \frac{b_3}{3} + \frac{b_5}{5} + \ldots\right) = 2\left(\frac{1}{1.2} + \frac{1}{3.4} + \frac{1}{5.6} + \ldots\right)$$

$$= 2\left(\frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \ldots\right) = 2\ln 2$$

17. (a) $\int_0^{2\pi} f(x)\cos kx\,dx = \sum_{r=0}^{n} \int_0^{2\pi} a_r \cos rx \cos kx\,dx$

$$= \frac{1}{2}\sum_{r=0}^{n} \int_0^{2\pi} a_r(\cos(r + k)x + \cos(r - k)x)\,dx$$

$$\frac{1}{2}\sum_{r=0}^{n}\left[a_r\left(\frac{\sin(r + k)x}{(r + k)} + \frac{\sin(r - k)x}{(r - k)}(r \neq k)\right)\right]_0^{2\pi} + \frac{1}{2}a_k \int_0^{2\pi} (\cos 2kx + 1)\,dx = \pi a_k$$

$$\Rightarrow \qquad \sum_{r=0}^{n} \int_0^{2\pi} f(x)\cos kx\,dx = \pi(a_0 + a_1 + a_2 + \ldots a_n = \pi f(0)$$

18. **(d)** $\int_0^{2\pi} b_r \sin rx \cos kx \, dx = \frac{1}{2} b_r \int_0^{2\pi} (\sin(r+k)x + \sin(r-k)x) \, dx = 0$

$\sum_{k=0}^{n} \int_0^{2\pi} b_r \sin rx \cos kx \, dx = 0$

SECTION–V
Matrix–Match Type

1. **(A) – 1, 4; (B) – 1, 3; (C) – 1, 3; (D) – 2, 4, 5**

(A) $f(x) = (x+1)^2 + x - [\sqrt{(x+1)^2 + (x+1)}]^2$

$(x+1)^2 < (x+1)(x+2) < (x+2)^2$ (as $x \in N$)

$(x+1) < \sqrt{(x+1)(x+2)} < (x+2)$

$\Rightarrow \quad f(x) = x$

So, $f(x)$ is one-one and into function

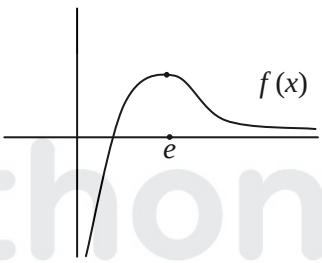
(B) $\frac{\log g(x)}{g(x)} = x$

$\Rightarrow \quad y = g(x)$ is inverse of $y = f(x)$

so its enough to investigate

$f : (0, 1) \to (-\infty, 0), \; y = f(x)$

which is one-one onto as



$f'(x) = \frac{1 - \log x}{x^2} = \frac{\log\left(\frac{e}{x}\right)}{x^2} > 0$

(C) $f : R \to R$ be continuous function

Such that $|f(x) - f(y)| \geq |x - y|$

$\Rightarrow \quad f(x) = f(y)$

If $x = y$

So f is one-one

Since $f(x)$ is continuous for all $x \in R$ and it is not bounded function.

$\Rightarrow \quad f(x)$ is onto function

(D) $f(x) = \int_0^x \frac{1}{1+t^2} dt + \int_0^{1/x} \frac{1}{1+t^2} dt$

$f(x) = \tan^{-1} x + \tan^{-1}\left(\frac{1}{x}\right)$

$f(x) = \frac{\pi}{2}$ for $x > 0$

$\frac{-\pi}{2}$ $x < 0$

So $f(x)$ is many one and into

2. (A) – 1, 2; (B) – 2, 3, 4 ; (C) – 4, 5; (D) – 5

(A) $|z_1|=|z_2|=|z_3|=1$

$z_1^3+z_2^3+z_3^3-3z_1z_2z_3=-4z_1z_3z_3$

or $(z_1+z_2+z_3)\left((z_1+z_2+z_3)^2-3(z_1z_2z_3)(\bar{z}_1+\bar{z}_2+\bar{z}_3)\right)=-4z_1z_2z_3$

or $(z_1+z_2+z_3)^3=z_1z_2z_3(3|z_1+z_2+z_3|^2-4)$

or $|z_1+z_2+z_3|^3=\left|3|z_1+z_2+z_3|^2-4\right|$ or $k^3=|3k^2-4|$

If $k>\dfrac{2}{\sqrt{3}}$

$k^3=3k^2-4$

$k^3-3k^2+4=0 \quad \Rightarrow \quad (k-2)^2(k+1)=0 \Rightarrow k=2$

If $0<k<\dfrac{2}{\sqrt{3}}$ $\quad k^3+3k^2-4=0 \Rightarrow (k-1)(k+2)^2=0 \Rightarrow k=1$

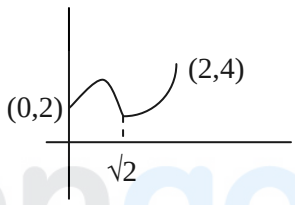
(B) Let $|1-z|=t$

So $0\le t\le 2$ or $z+\bar{z}=2-t^2$

Let $P=|1-z|+|1+z^2|$

$P=t+|z||z+\bar{z}|$

$P=t+|2-t^2|,\ 0\le t\le 2$



(C) $|1-\cos\theta-i\sin\theta|=|2\sin^2\dfrac{\theta}{2}-2i\sin\dfrac{\theta}{2}\cos\dfrac{\theta}{2}|=2|\sin\dfrac{\theta}{2}|$

So $k=\pm 2$

(D) $f(x)=x^4+ax^3+bx^2+cx+d$

As $\alpha, \beta, \gamma, \delta$ are real roots of equation $f(x)=0$

$\Rightarrow \quad f(x)=(x-\alpha)(x-\beta)(x-\gamma)(x-\delta)$

Given that $|f(i)|=1$

$\Rightarrow \quad (1+\alpha^2)(1+\beta^2)(1+\gamma^2)(1+\delta^2)=1$

It is only possible if $\alpha=\beta=\gamma=\delta=0$

So $a=b=c=d=0$